

Wireless AM Listening Bug

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The circuit presented here can be used to listen to the conversation going on in an adjacent room using an AM radio receiver.

Circuit and working

Circuit diagram of the wireless AM listening bug is shown in Fig. 1. It is built around microphone MIC1, two op-amp 741 ICs (IC1 and IC2), transistor SL100 (T1) and a few other

components.

The circuit works on 9V regulated power supply and is highly sensitive. Sound picked up by the microphone is amplified by IC1 and IC2.

Amplified audio signals are sent to a medium-wave RF oscillator.

Transmitting frequency of the circuit can be varied by adjusting variable capacitor VC1. Coil L1 is wound on plastic/paper former of 1cm dia having a length of around 6cm. The coil should have 90 turns of 36SWG

enameled copper wire, tapped at 75 turns from the antenna side.

A good-quality medium-wave AM radio is recommended to listen to the conversation. Range of the circuit is around 100m. Transistor SL100 can be replaced with 2N2219 or BC548.

After assembling the circuit, connect 9V supply to it. Sing a song in front of condenser microphone MIC1. Tune the AM radio until you receive the transmitted signal. If you are unable to receive the signal, tune variable capacitor VC1. Tune the AM radio until you receive the transmitted signal. For maximum range, use a sensitive radio with external wire antenna.

Construction and testing

An actual-size, single-side PCB for the wireless listening bug is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a small box in such a way that you can use it according to your requirement. Use

a 10m or longer piece of wire for the antenna that may have to be hidden from view. ●

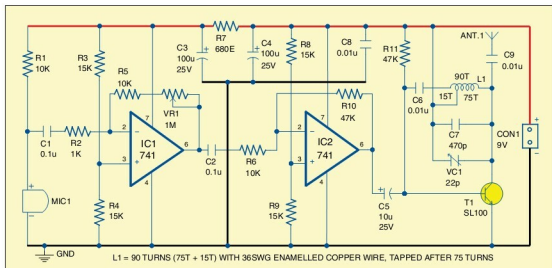


Fig. 1: Circuit diagram for wireless AM listening bug

PARTS LIST

Semiconductors:

- IC1, IC2 - 741 op-amp
- T1 - SL100 npn transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1, R5, R6 - 10-kilo-ohm
- R2 - 1-kilo-ohm
- R3, R4, R8, R9 - 15-kilo-ohm
- R7 - 680-ohm
- R10, R11 - 47-kilo-ohm
- VR1 - 1-mega-ohm potentiometer

Capacitors:

- C1, C2 - 0.1µF ceramic disk
- C3, C4 - 100µF, 25V electrolytic
- C5 - 10µF, 25V electrolytic
- C6, C8, C9 - 0.01µF ceramic disk
- C7 - 470pF ceramic disk
- VC1 - 22pF trimmer capacitor

Miscellaneous:

- CON1 - 2-pin connector terminal
- MIC1 - Condenser microphone
- L1 - 90 turns (75T + 15T) with 36SWG enameled copper wire, tapped after 75 turns
- ANT.1 - Wire antenna (10m or longer)

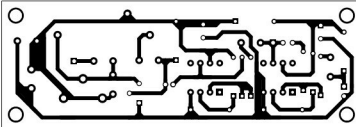


Fig. 2: Actual-size PCB pattern of the circuit for the wireless listening bug

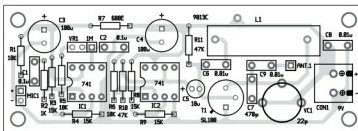


Fig. 3: Component layout of the PCB



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